Flight-First Cluster Design for Single-Pass Fixed-Wing UAV Query Delivery

Technical Report — Layer 2 of Phase 1 (flight line and clustering)
Version 22 Oct. 2026. All numbers are exploratory (one ns-3 pass per cluster; $N < 120$).

***Abstract*—A fixed-wing UAV delivers a search query (a clue set of Q packets) to a network of IoT cameras by flying over each cluster *once* and broadcasting one IEEE 802.15.4 packet every $0.5\,\mathrm{m}$. Because every packet is emitted at a known point of the path, the geometry of the flight line and of the clusters fixes how much data each cluster receives, which cells hear it in long contiguous runs, and how much ground cooperation (Layer 3, River) is left to do. This report presents the two Layer-2 procedures we built on this observation: CHORD, which flies the longest chord through the cluster head of a given cluster, and MEANDER, which partitions the land into River-wide lanes along its geodesic spine and flies a curvature-bounded path through each cluster head. CHORD raises the data per pass by up to 88 %; MEANDER raises the share of cells inside the River band from 55–57 % to 83–90 %. At mission level, however, MEANDER does not yet beat CHORD on existing clusters: it creates 20–65 % more clusters and a longer tour. We report why, the cluster shape at which River saturates (lanes 8–10 cells wide, 2–3 km long), and the fair comparison that remains to be run.**

I. Core Idea

Phase 1 of the mission has three layers: what a camera needs to recognise the target (Layer 1), how the UAV delivers it (Layer 2) and how the ground network completes the delivery (Layer 3). The UAV is fixed-wing: it cannot hover and passes over each cluster once, at $50\,\mathrm{m/s}$ and $100\,\mathrm{m}$. Cells near the path hear long runs of packets; cells far from it hear scattered packets that cannot be assembled into files. Layer 3 (River) then moves the missing packets to a few *critical cells* that will run the matching.

Key observation: in a single-pass broadcast, the geometry is the schedule. Packet s is sent at $\gamma(d_s)$, $d_s = d_0 + s\Delta$, $\Delta = 0.5\,\mathrm{m}$, and γ is planned by the base station (BS). Hence, before the flight,

- the in-cluster length of γ_k fixes the data D_k that cluster k receives;
- the cells within the River half-width ω of γ_k are those that hold long contiguous runs;
- the number of clusters fixes the number of entries, exits and turns of the tour, and the number of cluster heads (each one a critical cell).

Layer 2 should therefore design the flight lines and the clusters *together*. We pursue three principles: **(P1)** fly the longest chord through each cluster head; **(P2)** choose the flight lines first and cut the clusters along them (flight-first partitioning); **(P3)** size the clusters to the clue set and to the point where River saturates.

II. System Model and Layer-2 Problem

Domain. The land is a set $\mathcal{H}$ of hexagonal cells of radius $R = 100\,\mathrm{m}$ (width $w = \sqrt{3}R \approx 173\,\mathrm{m}$), about 21 nodes per cell. Lakes are holes: no nodes, but the UAV may fly over them. $d(a, b)$ is the geodesic distance over land cells. Each cell has a cell leader (CL), its strongest node.

UAV. Speed $v = 50\,\mathrm{m/s}$, minimum turn radius $\rho = 255\,\mathrm{m}$ (45° bank). One packet of $99\,\mathrm{B}$ payload every $10\,\mathrm{ms}$; the clue set has $Q = 4036$ packets, about $2.0\,\mathrm{km}$ of flight. Links between clusters are Dubins paths [1].

Outputs of Layer 2. Clusters $\{\mathcal{C}_k\}$ that partition $\mathcal{H}$, each connected with $|\mathcal{C}_k| \le \mathrm{Cap}$ ($\mathrm{Cap} = 80$); a cluster head CH_k, the strongest node at least $300\,\mathrm{m}$ from the cluster edge; a flight path γ_k through CH_k with curvature $\kappa \le 1/\rho$ and straight leads of length ρ outside $\mathcal{C}_k$; and a visiting order π.

Metrics. $D_k = \{s : \gamma(d_s) \in \mathcal{C}_k\}$; River coverage $\varrho = \sum_k |\{h \in \mathcal{C}_k : \mathrm{dist}(z_h, \gamma_k) \le \omega\}|/|\mathcal{H}|$ with $\omega = 1.75w \approx 303\,\mathrm{m}$; tour time T_1; and the mission time $T^\star$, at which the second UAV team has verified the top-scored cell $c^\star$. $T^\star$ couples all layers, so Layer 2 is designed on three proxies: $|D_k| \ge Q$ (the clue set fits), large ϱ (short River), and small T_1 (few clusters, short links).

III. Procedure 1: Longest Chord Through the Cluster Head

For an existing partition (capacitated geodesic k-medoids [2]), CHORD (Alg. 1) scans the straight lines through CH_k and flies the one with the longest in-cluster length. Entry and exit are the farthest points of the line still in the cluster, so the UAV crosses lakes and neighbour clusters in between (soft obstacles). The visiting order minimises the time at which every critical cell holds its data, solved exactly by Held–Karp [3].

On the four clusters of Fig. 1, the longest chord adds 61, 5, 88 and 79 % of in-cluster length. On all 200 clusters of our 40 evaluation maps, the head margin had to drop to $200\,\mathrm{m}$ for 14 thin clusters, and the longest chord was always flyable: the loop of lines 5–8 never needed a second direction.

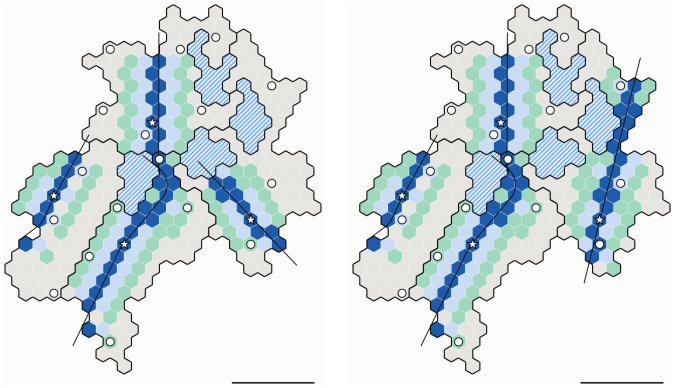
IV. Procedure 2: Flight-First Partitioning (MEANDER)

CHORD keeps clusters that were cut without regard to the flight: a compact cluster has only about half of its cells within ω of any single line. MEANDER (Alg. 2) reverses the order. It first lays out lanes as wide as the River, following the shape of

Algorithm 1 CHORD: flight line of one cluster

Input: cluster $\mathcal{C}$, its nodes, radius ρ, steps $\delta\theta = 2^\circ$, $\delta t = 5\,\mathrm{m}$
Output: CH, entry p_{in}, exit p_{out}, path γ

1: **for** margin $m \in (300, 200, 100, 0)\,\mathrm{m}$ **do**
2: CH $\leftarrow$ strongest node with edge distance $\geq m$; **if** found **break**
3: **for** $\theta \in \{0, \delta\theta, \ldots, 180^\circ - \delta\theta\}$ **do** ▷ θ, $\theta + \pi$: one line
4: $L(\theta) \leftarrow \delta t \cdot |\{t \in \delta t\mathbb{Z} : |t| \leq 8\,\mathrm{km},\ \mathrm{cell}(\mathrm{CH} + t\,u_\theta) \in \mathcal{C}\}|$
5: **for** θ in decreasing order of $L(\theta)$ **do**
6: $p_{\mathrm{in}}, p_{\mathrm{out}} \leftarrow$ farthest points of the line in $\mathcal{C}$, one on each side of CH
7: $\gamma \leftarrow$ shortest Dubins path $p_{\mathrm{in}} \rightarrow \mathrm{CH} \rightarrow p_{\mathrm{out}}$ (radius ρ) whose end headings cross the edge and whose leads of length ρ lie outside $\mathcal{C}$
8: **if** $\gamma \neq \emptyset$ **return** $(\mathrm{CH}, p_{\mathrm{in}}, p_{\mathrm{out}}, \gamma)$



(a) (b)

Fig. 1. CHORD on a four-cluster map (bar: 1 km). Dark blue: Axe (cells under the path); light blue: River; green: Bank; star: CH; circle: other critical cell; hatched: lake. (a) Lines through CH from a random edge point. (b) The east cluster now flies its longest chord, across the lake: 1.25 km → 2.34 km in the cluster, $|D|$ 2494 → 4685 packets (+88 %).

the land, and then cuts each lane into clusters along its length. The steps are shown in Fig. 2.

Geodesic coordinates (lines 1–4). The spine is the shortest land path between the two ends of the geodesic diameter, found with two Dijkstra sweeps [4]. The coordinate $s(h) = (d(A,h) - d(B,h))/2$ runs along the land and bends round lakes; $n(h)$ is the signed geodesic distance to the spine, its side taken from the spine's local tangent where the wavefront leaves it. Iso-lines of n follow the bends where straight lanes [5], [6] would cut across them.

Lanes and clusters (lines 5–10). Lanes are strips of n of width Lw. Three offsets are tried and the one with the fewest clusters plus fragments is kept. Lakes split lanes into pieces; pieces smaller than $f \cdot \mathrm{Cap}$ join the neighbour with the longest shared border. Each piece is cut along s into $\lceil |P|/\mathrm{Cap} \rceil$ equal consecutive chunks, so every cluster is long and narrow and its path runs along it.

Curvature-bounded path through the head (Alg. 3). The centre curve of a lane cluster is smoothed by a Laplacian iteration until $\kappa \leq 1/\rho$. It must also pass through CH_k, which hears the UAV directly and is always critical. BEND adds a Gaussian bump $v\,e^{-(s-s^\star)^2/2\sigma^2}$ towards the head. Its peak

Algorithm 2 MEANDER: flight-first partitioning (BS)

Input: land cells $\mathcal{H}$ (lakes are holes), Cap, lane width L, fragment ratio f, radius ρ
Output: clusters $\{\mathcal{C}_k\}$, paths $\{\gamma_k\}$, order π

1: **for** each connected land piece **do**
2: $A \leftarrow \arg\max_h d(h_0, h)$; $B \leftarrow \arg\max_h d(A, h)$; Spine $\leftarrow$ path $A \rightarrow B$
3: $s(h) \leftarrow \big(d(A,h) - d(B,h)\big)/2$ ▷ along
4: $n(h) \leftarrow$ signed geodesic distance to Spine ▷ across
5: $o^\star \leftarrow \arg\min_{o \in \{0, \frac{1}{3}, \frac{2}{3}\}}$ #clusters + #fragments of $\mathrm{lane}_o(h) = \lfloor n(h)/(Lw) + o \rfloor$
6: $\mathcal{P} \leftarrow$ connected pieces of the lanes $\mathrm{lane}_{o^\star}$
7: **while** some $P \in \mathcal{P}$ with $|P| < f\mathrm{Cap}$ touches another piece **do**
8: merge the smallest such P into its longest-border neighbour
9: **for** each $P \in \mathcal{P}$ **do**
10: sort P by s; cut into $\lceil |P|/\mathrm{Cap} \rceil$ equal consecutive chunks
11: **for** each cluster $\mathcal{C}_k$ **do**
12: $\mathrm{CH}_k \leftarrow$ as in Alg. 1, line 2; $\gamma_k \leftarrow \mathrm{BEND}(\mathcal{C}_k, s, \mathrm{CH}_k, \rho)$
13: $\pi \leftarrow$ shortest of 4 boustrophedon tours (lane by lane, alternating)

Algorithm 3 BEND: path of a lane cluster through its head

Input: cluster $\mathcal{C}$, coordinate s, head CH, radius ρ

1: $P \leftarrow$ mean cell centre per $w/2$-bin of s; moving average over 7 bins
2: SMOOTH(P): $P_i \leftarrow P_i + \frac{1}{4}(P_{i-1} + P_{i+1} - 2P_i)$ until $\max \kappa \leq 1/\rho$
3: $s^\star \leftarrow \arg\min_s \|P(s) - \mathrm{CH}\|$; $v \leftarrow \mathrm{CH} - P(s^\star)$
4: **for** $c \in (1, 1.5, 2, 3, 4, 6)$ **do**
5: $\sigma \leftarrow c\sqrt{2|v|\rho}$; $Q(s) \leftarrow P(s) + v\,e^{-(s-s^\star)^2/(2\sigma^2)}$
6: SMOOTH(Q) with CH and both ends pinned; **if** $\max \kappa \leq 1/\rho$ **break**
7: **return** lead-in of length $\rho \oplus Q \oplus$ lead-out of length ρ

curvature is about $|v|/\sigma^2$, so $\sigma = \sqrt{2|v|\rho}$ keeps the bump alone within $1/(2\rho)$; the curve is then smoothed again with the head pinned (Fig. 3). All 28 clusters of three mission maps were bent successfully; one needed σ tripled. On a curved Axe, Layer 3 builds the Banks along the *local* normal of γ: at a 70° exit angle this raises River coverage from 0.78 to 0.98 and Bank offset from the path from 0.63ω to 0.88ω (ideal: ω), with unchanged ns-3 completion times (9.0 vs. 9.1 s on average).

V. RESULTS

Setup. Geometry: 57 curved domains (90°, 180° or 270° bends, 0.6 or 1 km wide, 4 or 6 clusters' worth of land). Mission: 40 "lake-block" maps with $K = 2, 4, 6, 8$ clusters (10 each), BS 1 km south; the pass and River run in ns-3 (IEEE 802.15.4 PHY and full MAC [7], urban A2G and G2G channels); the second team is replayed for 100 random draws per map. Baseline: k-medoids clusters flown by CHORD.

Coverage (Table I, Fig. 4). Flight-first partitioning puts 83–90 % of the cells within the River band, against 55–57 % for k-medoids with CHORD, and halves the distance of the farthest cell from its path. On the 40 mission maps it doubles the share of cells that hold all 12 clue images on their own, without River (17–28 % vs. 10–14 %). Against straight lanes, the geodesic lanes gain most in narrow bent strips (+8 points

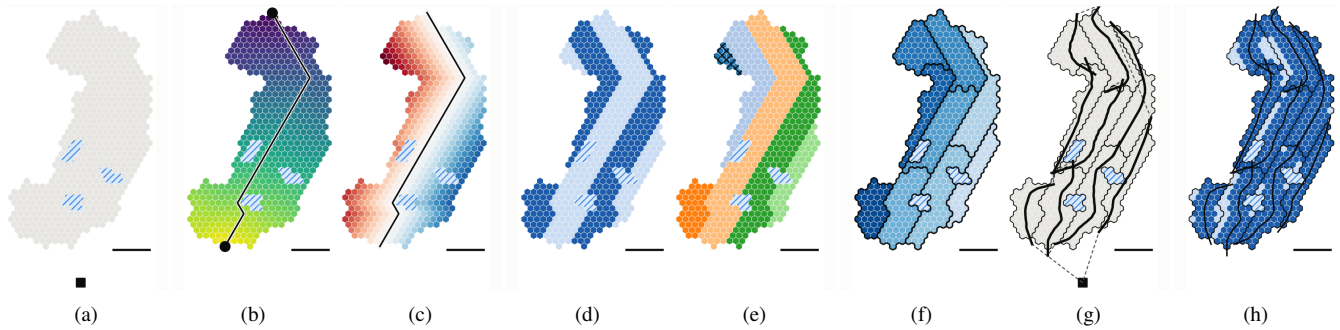


Fig. 2. MEANDER step by step on a curved domain (220° bend, about 1 km wide, 480 cells, three lakes; bar: 1 km). (a) Land, lakes (hatched) and BS (square). (b) Spine $A \to B$ (geodesic diameter) and along-coordinate s, purple at A to yellow at B. (c) Across-coordinate n: signed geodesic distance to the spine, red on one side, blue on the other. (d) Lanes of width Lw, alternating shades. (e) Lane pieces; the hatched fragments are merged into a neighbour. (f) Clusters, darker = visited earlier. (g) Smoothed flight curves (bold) and the tour (dashed). (h) River coverage: dark = within ω of the own cluster's path (88.5 % of cells).

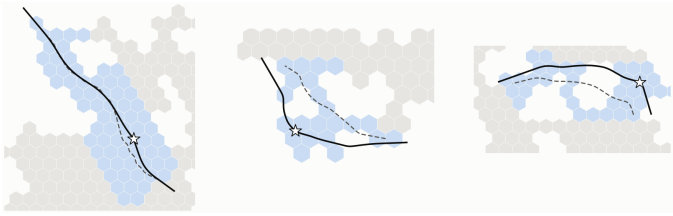


Fig. 3. BEND on three mission clusters: a small offset; a 370 m offset (L-shaped cluster); a 302 m offset (U-shaped cluster round a lake). Dashed: smoothed centre curve; solid: final path through the head (star), with its leads.

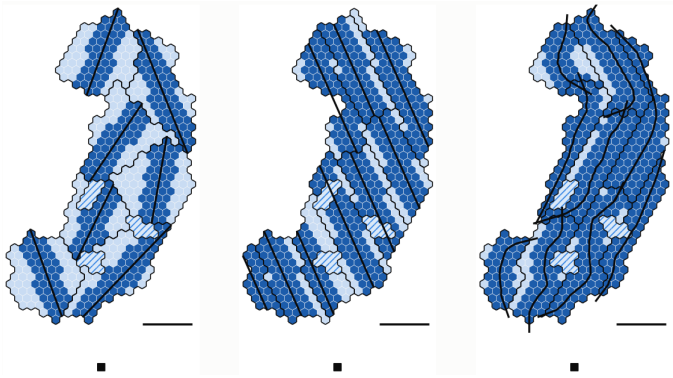


Fig. 4. One curved domain, three partitions: k-medoids with CHORD (left), straight lanes (middle), MEANDER (right). Dark blue: cells within ω of the own cluster's path; black: paths. River coverage 57, 80 and 88 %.

TABLE I
GEOMETRY ON 57 CURVED DOMAINS (RANGE OVER SIX DOMAIN CLASSES; DISTANCES IN CELLS UNLESS NOTED)

	k-med. + CHORD	Straight lanes	MEANDER
Clusters	5.2–7.8	7.2–9.3	7.1–9.3
River coverage ϱ	0.55–0.57	0.81–0.87	**0.83–0.90**
Farthest cell from path	4.8–5.3	2.2–2.4	2.2–2.8
In-cluster flight (km)	11.0–15.3	14.3–20.5	17.1–27.1

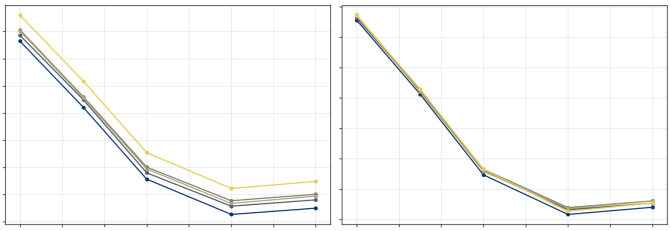


Fig. 5. Domain-level time at which critical cells hold their data on a 6×4 km domain, versus lane width (3, 4.5, 6, 8, 10 cells). Left: mean; right: latest. One line per cluster length, dark blue (1.5 km) to yellow (6 km). The minimum lies at 8 cells.

at 0.6 km and 90°, 10 of 10 domains) and 0–3 points on wide land.

Where River saturates (Fig. 5). We swept strip-shaped clusters of width $W \in \{3, \ldots, 10\}$ cells and length $L \in \{1.5, \ldots, 6\}$ km (75 ns-3 runs). (i) *Length:* with one 400 kB window around the head, the last critical cell completes 2.7–4.8× later at 3 km than at 2 km (53–102 s). Giving every critical cell the copy of the clue set nearest to it removes this limit: 19–32 s even at 6 km. Below 2 km no copy fits. (ii) *Width:* completion stays at 7–10 s up to about 600 m from the path; the network load per cell grows slowly up to 8 cells (227→385 frames), then 1.6× at 10 cells and another 2.1× at 14 cells. (iii) *Domain level:* the tour dominates, since each lane fewer saves 100–150 s. The recommended shape is lanes 8–10 cells wide, clusters 2–3 km long (110–190 cells): Cap = 80 is too small for lane clusters.

Mission level (Table II, Fig. 6). The coverage gain does not yet turn into a faster mission. With Dubins links, MEANDER (tuned: $L = 7.5$, $f = \frac{1}{2}$) finishes 48–97 s later than k-medoids with CHORD and wins on at most 2 of 10 maps per size. Three effects explain it:

- *More clusters* (+20–45 % after tuning): more entries, exits and Dubins turns, and more cluster heads, hence more critical cells for the second team to rank.
- *Raw data already penetrates.* The A2G link reaches about 1 km, so 98–100 % of cells hear at least half of their cluster's data under any partition; River completes the clue set at the critical cells in tens of seconds.
- *Same verification once $c^\star$ is known.* With an oracle for $c^\star$ both partitions finish together (244/287/316/345 s

TABLE II
MISSION LEVEL ON 40 MAPS (DUBINS LINKS, 100 DRAWS PER MAP)

	Partition	$K = 2$	$K = 4$	$K = 6$	$K = 8$
Clusters	k-med. + CHORD	2	4	6	8
	MEANDER	3.3	6.2	9.0	12.1
	MEANDER, tuned	2.9	5.1	7.2	9.7
Critical cells	k-med. + CHORD	13.5	26.9	39.9	50.1
	MEANDER	16.5	30.6	47.3	59.0
	MEANDER, tuned	15.5	29.5	42.8	55.5
$T^\star$ (s)	k-med. + CHORD	**420**	**586**	**716**	**824**
	MEANDER	487	679	845	987
	MEANDER, tuned	479	634	770	921
Maps won (of 10)	MEANDER	1	0	0	0
	MEANDER, tuned	1	1	0	2

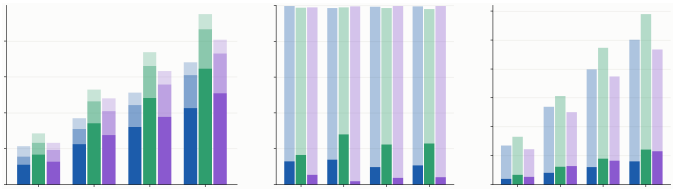


Fig. 6. Diagnosis on the 40 maps, groups $K = 2, 4, 6, 8$; bars: k-medoids + CHORD (blue), MEANDER (green), split-cluster baseline (purple). Left: first-team tour, stacked in-cluster (dark), links (medium), return (light). Middle: cells hearing $\geq 50\,\%$ of D (light) and cells holding all 12 clues alone (dark). Right: critical cells (light) and clusters (dark).

vs. 245/287/314/347 s); the loss lies in the longer tour (reports arrive later) and in the larger candidate set.

VI. DISCUSSION AND NEXT STEPS

What holds. (i) Flight geometry is a design variable of the network: CHORD alone nearly doubles the data of a cluster without adding any cluster. (ii) Geodesic lanes give near-complete River coverage on curved, perforated land with curvature-feasible paths. (iii) River's cost depends on the cluster shape in a predictable way, which gives concrete targets for Layer 2.

What does not, yet. Coverage is not the objective: critical cells need only the clue set, and River supplies it quickly. Cluster count and tour length dominate $T^\star$. The current MEANDER also pays costs that do not belong to the lane idea: lanes cut into fragments by lakes, and a separate path per cluster instead of one continuous sweep. Kinematics penalise it further, since every cluster boundary is a turn (gap to k-medoids 62–150 s with straight links, 67–163 s with Dubins links).

Fair comparison (pending). The partitions above differ in cluster number and size (coefficient of variation 0.16–0.21 vs. 0.06–0.11). MEANDER-K chains all lanes into one boustrophedon sequence that flies over lakes, cuts it into exactly K equal runs, and keeps the paths, heads and River of MEANDER; it should be evaluated at the saturation optimum (lanes about 8 cells) with per-critical-cell windows for both partitions.

Use in the ICC paper. Sections II–III (model, CHORD) fit the system-model part; Section IV and Fig. 2 can form a "flight-first clustering" subsection; Tables I–II and Fig. 5 support a short Layer-2 evaluation once the fair comparison is run with $N \geq 120$.

REFERENCES

[1] L. E. Dubins, "On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents," *Amer. J. Math.*, vol. 79, no. 3, pp. 497–516, 1957.

[2] L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*. New York: Wiley, 1990.

[3] M. Held and R. M. Karp, "A dynamic programming approach to sequencing problems," *J. SIAM*, vol. 10, no. 1, pp. 196–210, 1962.

[4] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numer. Math.*, vol. 1, pp. 269–271, 1959.

[5] W. H. Huang, "Optimal line-sweep-based decompositions for coverage algorithms," in *Proc. IEEE ICRA*, 2001, pp. 27–32.

[6] H. Choset, "Coverage of known spaces: The boustrophedon cellular decomposition," *Auton. Robots*, vol. 9, no. 3, pp. 247–253, 2000.

[7] *IEEE Standard for Low-Rate Wireless Networks*, IEEE Std 802.15.4-2020, 2020.